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Question:
The X coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as $(0, 1)$, $(1, 1)$ and $(1, 0)$ is (2013)

Solution:

Let the position vectors of the midpoints of sides BC, AC, and AB be $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{F}$ respectively.

$$\mathbf{D} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{E} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{1}$$

The position vectors of the vertices of the triangle, $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, can be found from the midpoint vectors.

$$\mathbf{A} = \mathbf{E} + \mathbf{F} - \mathbf{D} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \tag{2}$$

$$\mathbf{B} = \mathbf{D} + \mathbf{F} - \mathbf{E} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{3}$$

$$\mathbf{C} = \mathbf{D} + \mathbf{E} - \mathbf{F} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \tag{4}$$

Next, we find the lengths of the sides of $\triangle ABC$. Let a, b, and c be the lengths of the sides opposite to vertices A, B, and C.

$$a = \|\mathbf{C} - \mathbf{B}\| = \left\|\begin{pmatrix} 0 \\ 2 \end{pmatrix}\right\| = \sqrt{0^2 + 2^2} = 2 \tag{5}$$

$$b = \|\mathbf{A} - \mathbf{C}\| = \left\|\begin{pmatrix} 2 \\ -2 \end{pmatrix}\right\| = \sqrt{2^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2} \tag{6}$$

$$c = \|\mathbf{B} - \mathbf{A}\| = \left\|\begin{pmatrix} -2 \\ 0 \end{pmatrix}\right\| = \sqrt{(-2)^2 + 0^2} = 2 \tag{7}$$

The position vector of the incentre, $\mathbf{I}$, is given by the formula:

$$\mathbf{I} = \frac{a\mathbf{A} + b\mathbf{B} + c\mathbf{C}}{a + b + c} \tag{8}$$

Substituting the vectors and side lengths:

$$\mathbf{I} = \frac{2\begin{pmatrix}2\\0\end{pmatrix} + 2\sqrt{2}\begin{pmatrix}0\\0\end{pmatrix} + 2\begin{pmatrix}0\\2\end{pmatrix}}{2 + 2\sqrt{2} + 2} \tag{9}$$

$$= \frac{1}{4 + 2\sqrt{2}}\left(\begin{pmatrix}4\\0\end{pmatrix} + \begin{pmatrix}0\\0\end{pmatrix} + \begin{pmatrix}0\\4\end{pmatrix}\right) \tag{10}$$

$$= \frac{1}{4 + 2\sqrt{2}}\begin{pmatrix}4\\4\end{pmatrix} = \begin{pmatrix}\frac{4}{4+2\sqrt{2}}\\ \frac{4}{4+2\sqrt{2}}\end{pmatrix} \tag{11}$$

The x-coordinate of the incentre, I_x, is the first component of $\mathbf{I}$.

$$I_x = \frac{4}{4 + 2\sqrt{2}} = \frac{2}{2 + \sqrt{2}} \tag{12}$$

$$= \frac{2(2 - \sqrt{2})}{(2 + \sqrt{2})(2 - \sqrt{2})} = \frac{2(2 - \sqrt{2})}{4 - 2} = 2 - \sqrt{2} \tag{13}$$

$$\implies I_x = 2 - \sqrt{2} \tag{14}$$